Lecture 7 Notes

Quantitative Data Analysis

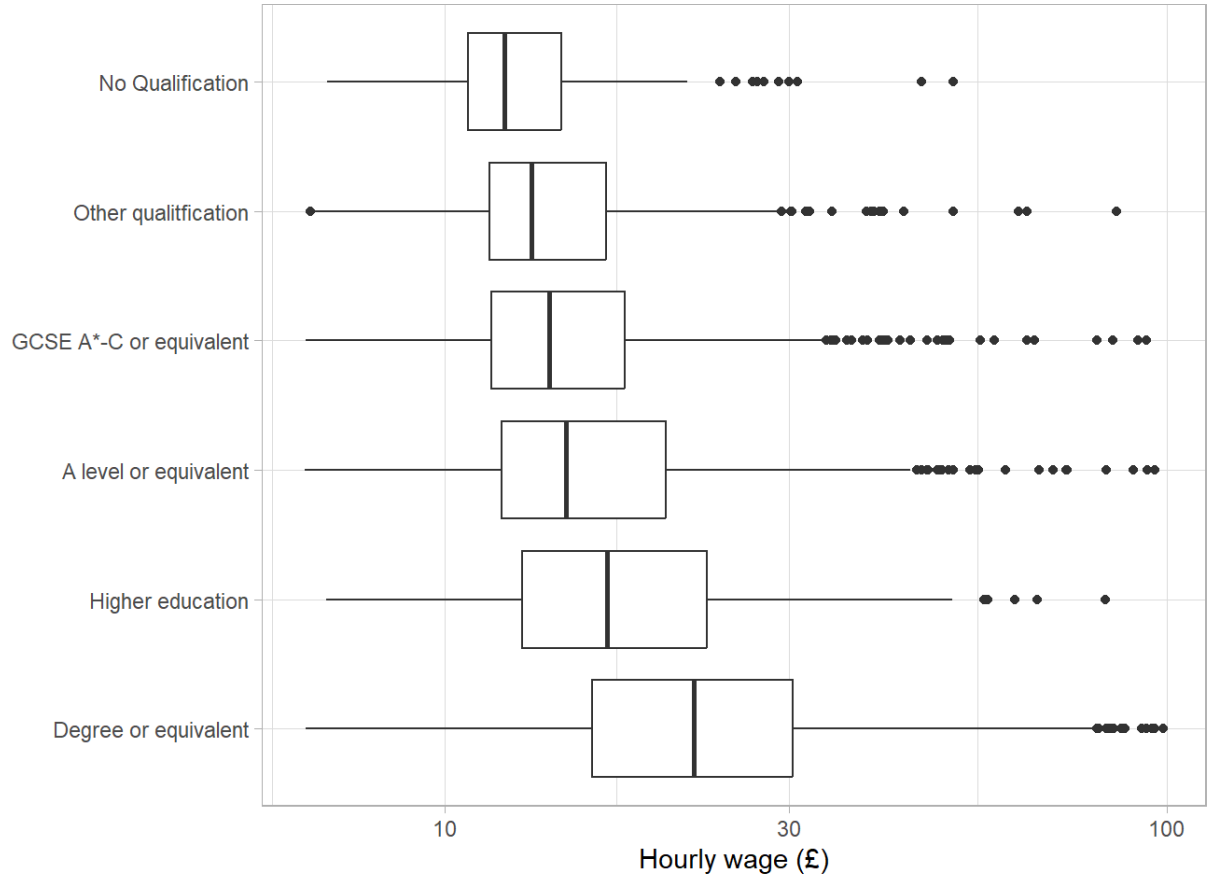
Introduction

- Last time, we met the idea of testing whether observed differences were due to sampling error
- We tested for differences between average male and average female wages
- Today we will learn how to test for differences between more than two categories
- Let's see whether the average wage varies by education level (measured as the highest qualification a person has obtained)
- For all our sake, let's hope so!

Level of highest qualification

Highest Qualification	Obs.	Mean Wage (£)	Std. Dev.
Degree or equivalent	3535	25.02	12.65
Higher education	565	19.63	9.88
A level or equivalent	1604	17.33	8.86
GCSE A*-C or equivalent	1102	16.16	8.29
Other qualitfication	331	15.63	8.55
No Qualification	210	13.53	5.39

Level of highest qualification



Education

- We can see from the chart that the median wage varies with education
- There is, however, substantial overlap
- If we drew a new sample, would the differences be reproduced

What is our hypothesis?

- We think education and wages are related

$$H_0 : \mu_{degree} = \mu_{HE} = \mu_{Alevel} = \mu_{GCSE} = \mu_{Other} = \mu_{None}$$

$$H_A : \mu_{degree} \neq \mu_{HE} \neq \mu_{Alevel} \neq \mu_{GCSE} \neq \mu_{Other} \neq \mu_{None}$$

The linear model

- We can use framework of the linear model to write down an equation
- Define a set of dummy variables which take the value if:
    - One if the qualification in question is the highest the person possesses
    - Zero otherwise

The regression equation

$$\text{Wage}_i = \beta_0 + \beta_1 \cdot \text{HE}_i + \beta_2 \cdot \text{A level}_i + \beta_3 \cdot \text{GCSE}_i + \beta_4 \cdot \text{Other}_i + \beta_5 \cdot \text{None}_i$$

The parameters

- β_0 will be the average wage of a person with a degree
- β_1 is the difference between the average wage of a person with a degree and a person with only some higher education
- β_2 is the difference between the average wage of a person with a degree and a person with A levels
- β_3 is the difference between the average wage of a person with a degree and a person with GCSEs
- β_4 is the difference between the average wage of a person with a degree and a person with some other qualification
- β_5 is the difference between the average wage of a person with a degree and a person with no qualifications

The linear regression model

```
lm(HOURPAY ~ HIQUL22D, data = lfs) %>% summary()
```

The linear regression model

```
##
## Call:
## lm(formula = HOURPAY ~ HIQUL22D, data = lfs)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -18.595  -6.305  -2.448   3.576  78.822
##
## Coefficients:
##                                Estimate Std. Error t value Pr(>|t|)
## (Intercept)                     25.0155     0.1812  138.06   <2e-16 ***
## HIQUL22DHigher education        -5.3873     0.4881  -11.04   <2e-16 ***
## HIQUL22DA level or equivalent   -7.6878     0.3243  -23.70   <2e-16 ***
## HIQUL22DGCSE A*-C or equivalent -8.8511     0.3717  -23.81   <2e-16 ***
## HIQUL22DOther qualitfication    -9.3824     0.6192  -15.15   <2e-16 ***
## HIQUL22DNo Qualification       -11.4876     0.7652  -15.01   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 10.77 on 7341 degrees of freedom
## Multiple R-squared:  0.1293, Adjusted R-squared:  0.1287
## F-statistic: 218.1 on 5 and 7341 DF,  p-value: < 2.2e-16
```

Interpretation

- We have some strong results here
- Every level of education below degree has a lower wage on average
- All of the differences are statistically significant
- In this case, we have enough evidence to reject the null hypothesis we set up earlier
- However, this might not always be the case
- We may find instances where nothing is individually significant, but the parameters are *jointly* significant

Variation

- Variation in the data can occur along two dimensions
    1. Within a group (within an education category)
    2. Between groups (between education categories)
- The question we need to answer is whether the variation between groups is significantly greater than the variation from person to person

Formalising these concepts

- We need to formalise the concepts of variation we have discussed:
    1. Total variation: the *sum of squares total* (SST)
    2. Among-group variation: *sum of squares among groups* (SSA)
    3. Within-group variation: *sum of squares within groups* (SSW)

Total variation

We can define the sum of squares total as:

$$SST = \sum_{j=1}^{c} \sum_{i=1}^{n_j} (X_{ij} - \bar{\bar{X}})^2$$

where

- $\bar{\bar{X}} = \frac{\sum_{j=1}^{c} \sum_{i=1}^{n_j} X_{ij}}{n}$ = Grand Mean
- X_{ij} is the ith value in group j
- n_j is the number of values in group j
- n is the total number of values across all groups
- c is the number of groups

Among-group variation

We define the among group variation as:

$$SSA = \sum_{j=1}^{c} n_j (\bar{X}_j - \bar{\bar{X}})^2$$

where

- c is the number of groups
- n_j is the number of values in group j
- $\bar{X}_j$ is the sample mean of group j
- $\bar{\bar{X}}$ is the grand mean

Within-group variation

We define the within-group variation as:

$$SSW = \sum_{j=1}^{c} \sum_{i=1}^{n_j} (X_{ij} - \bar{X}_j)^2$$

where

- X_{ij} is the ith value in group j
- $\bar{X}_j$ is the sample mean of group j

Mean squares

- We can convert these three sums of squares (SS) into variances or mean squares by dividing them by their degrees of freedom (d.o.f.)

    1. The among-group SS compares c groups and contributes $c - 1$ d.o.f.
    2. Since each of the c groups contributes $n_j - 1$ d.o.f. , there are $n - c$ d.o.f. associated with the within-group SS
    3. The total variation contributes $n - 1$ d.o.f. since we have n values and 1 grand mean

The Mean Square Errors

The Mean Square Among:

$$MSA = \frac{SSA}{c - 1}$$

The Mean Square Within:

$$MSW = \frac{SSW}{n - c}$$

The Mean Square Total:

$$MST = \frac{SST}{n - 1}$$

Testing

- To conduct our test we need to:

    1. State our null and alternative hypotheses
    2. Define our test statistic
    3. Choose a level of significance

Hypotheses

$$H_0 : \mu_1 = \mu_2 = \ldots = \mu_c$$

$$H_A : \mu_1 \neq \mu_2 \neq \ldots \neq \mu_c$$

- The null hypothesis states that the means for each group are not statistically different from each other
- The alternative states that not all of the means are equal
- It may be that only 1 mean is statistically significantly different from the others
- The method will not tell us which mean(s) is(are) different

The test statistic

- To test our null hypothesis, we use an F-test
- This makes use of an F-distribution

$$F = \frac{MSA}{MSW}$$

- F is our test statistic with $c - 1$ degrees of freedom on the numerator and $n - c$ degrees of freedom on the denominator
- MSA is the Mean Square Among
- MSW is the Mean Square Within

The F-test

- Mercifully we don't have to calculate it all by hand
- We can ask R to do it using the analysis of variance (ANOVA) function

```
aov(HOURPAY ~ HIQUL22D, data = lfs) %>% summary()
```

```
##               Df Sum Sq Mean Sq F value Pr(>F)
## HIQUL22D       5 126542   25308   218.1 <2e-16 ***
## Residuals   7341 851935     116
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Interpretation

- You can see that R gives us some details of the underlying calculations
- It tells us the final F value
- Crucially we get the p-value
- It is well below zero so we can safely reject our null hypothesis
- It seems the average wage varies with education level

Additional information

- Note that the *Sum Sq* column presents the sum of squares
- This is a measure of the variation in the data
- It is broken into two components
    1. Variation attributable to our categorical variable (HIQUL22D)
    2. Variation which we have failed to explain (Residuals)
- From this we can calculate how much variation is explained by our education variable
- We divide the sum of squares from HIQUL22D by the total sum of squares

$$\frac{126542}{126542 + 851935} = 0.129325$$

Interpretation

- This value comes to 0.129325
- It indicates that education levels explain about 12.93% of the variation in wage levels
- With all this in mind, let us return to our linear model from the start of the lecture

It was there all along

```
Call:
lm(formula = HOURPAY ~ HIQUL22D, data = lfs)

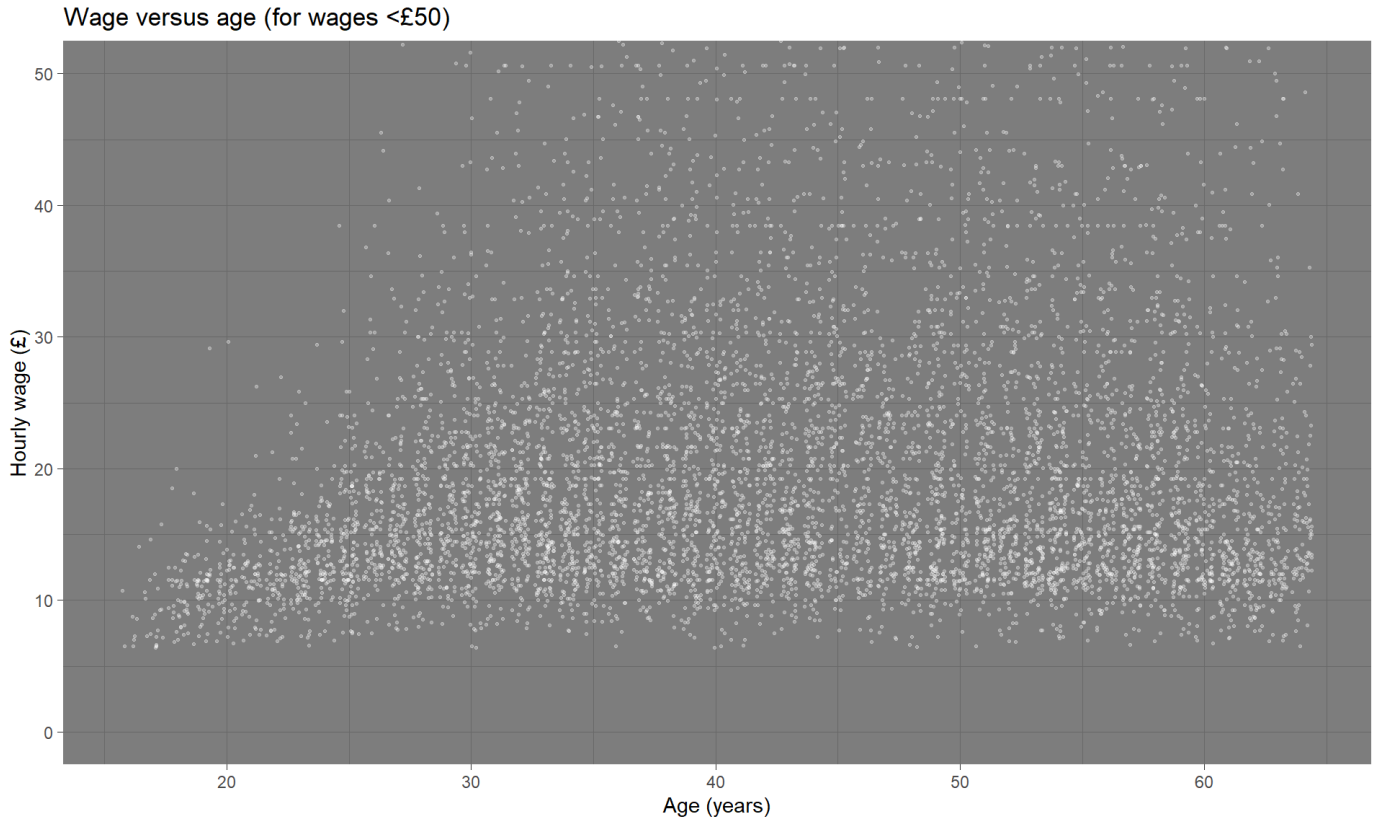
Residuals:
    Min      1Q  Median      3Q     Max
-18.595  -6.305  -2.448   3.576  78.822

Coefficients:
                              Estimate Std. Error t value Pr(>|t|)
(Intercept)                    25.0155     0.1812  138.06   <2e-16 ***
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Residual standard error: 10.77 on 7341 degrees of freedom
Multiple R-squared:  0.1293,    Adjusted R-squared:  0.1287
F-statistic: 218.1 on 5 and 7341 DF,  p-value: < 2.2e-16
```

The linear model

- We have now seen some of the magic of the linear model
- An assumption of the model is that the dependent variable is interval/ratio
- We have seen how categorical variables can be included as independent variables
- Ordinal variables work in the same way, expect that the model doesn't utilise their ordinal nature
- We can also have an interval/ratio variable as an independent variable
- Suppose we want to know about age and earnings



Measuring a relationship

- Visualising the data is an important step in understanding the relationship between two variables
- It is also useful to be able to quantify the relationship between two variables
- This step will also be necessary if we want to test for statistical significance
- There are several ways we can do this

Variance

- We have met the idea of variance in reference to one variable
- We can look at how two variables co-vary
- Recall that the sample variance is given as:

$$S_x^2 = \frac{\sum_i^n (x_i - \bar{x})^2}{n-1}$$

- The sample covariance is given by:

$$\text{cov}(X, Y) = \sum_{i=1}^{N} \frac{(x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$

Covariance

- If when x is above its mean, y is above its mean, the covariance is positive
- If when x is below its mean, y is above its mean, the covariance is negative
- If x moving tells us nothing about what is happening with y, the covariance is zero
- The units are hard to interpret because they depend on the units of x and y
- A standard approach is to normalise it by the product of the individual variances
- This gives the so-called *Pearson product moment correlation coefficient*, a measure bound between -1 and 1

$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}}$$

Interpreting the absolute value of r

Range	Relative strength
0	No relationship
0-0.2	Very weak, negligible relationship
0.2-0.4	Weak, low association
0.4-0.7	Moderate association
0.7-0.9	Strong, high, marked association
0.9-1	Very high, very strong relationship
1	Perfect association

Getting a correlatoin coefficient in R

- To get just the correlation coefficient we can use the *cor()* function

```
cor(lfs$HOURPAY, lfs$AGE)
```

```
## [1] 0.1375462
```

Testing statistical significance

- We can use the *cor.test()* function to test the statistical significance of the correlation

```
cor.test(lfs$HOURPAY, lfs$AGE)
```

```
##
##  Pearson's product-moment correlation
##
## data:  lfs$HOURPAY and lfs$AGE
## t = 11.901, df = 7345, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.1150412 0.1599101
## sample estimates:
##       cor
## 0.1375462
```

Testing statistical significance

- We can get the same thing using formula notation

```
cor.test(~ HOURPAY + AGE, data=lfs)
```

```
##
##  Pearson's product-moment correlation
##
## data:  HOURPAY and AGE
## t = 11.901, df = 7345, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.1150412 0.1599101
## sample estimates:
##       cor
## 0.1375462
```

The linear model

```
lm(HOURPAY ~ AGE, data=lfs) %>% summary()
```

```
##
## Call:
## lm(formula = HOURPAY ~ AGE, data = lfs)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -17.155  -7.615  -3.334   4.221  76.885
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 15.18591    0.49378   30.75   <2e-16 ***
## AGE          0.13233    0.01112   11.90   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 11.43 on 7345 degrees of freedom
## Multiple R-squared:  0.01892,    Adjusted R-squared:  0.01879
## F-statistic: 141.6 on 1 and 7345 DF,  p-value: < 2.2e-16
```

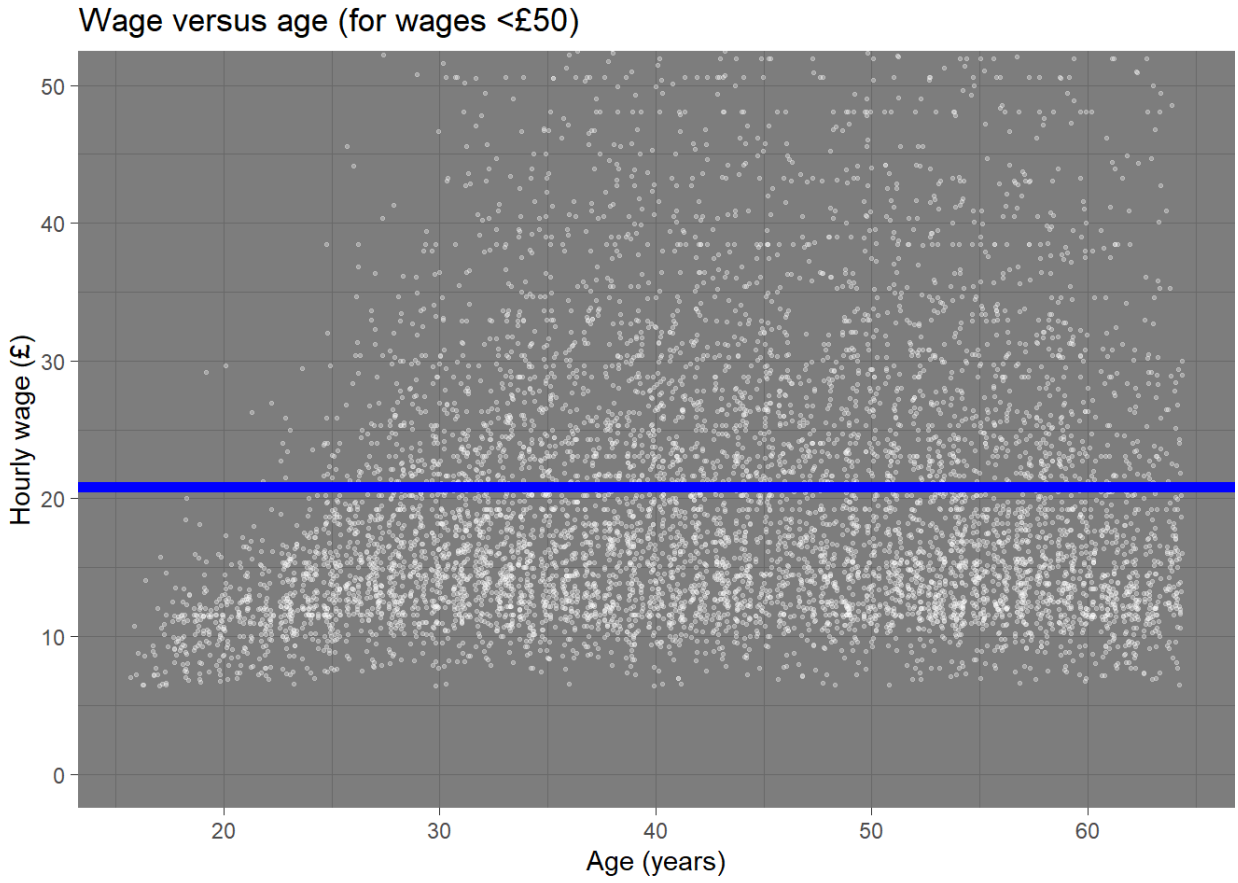
The linear model

- We can't see the correlation coefficient directly here, but it is here
- If we take the square root of the R^2, then we get the correlation coefficient
- In this case, it happens that the *Estimate* is also close to the correlation coefficient
- This is a coincidence, although the two measures are related
- Let's first find out where it comes from

Guessing a person's wage

- If I draw a random person from the population and ask you to guess their wage, your best guess will be the mean

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## ℹ Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



A better guess

- We can see that younger people have lower wages
- If we knew a person's age, we could make a better prediction of their wage
- Currently, our guess is given by the equation $\text{HOURPAY}_i = \beta_0$
- Let's add age into our equation $\text{HOURPAY}_i = \beta_0 + \beta_1 \cdot \text{AGE}$
- This isn't quite right, because we can't pick values for β_0 and β_1 which would perfectly predict wages
- Let's add a term to capture our error: $\text{HOURPAY}_i = \beta_0 + \beta_1 \cdot \text{AGE} + \varepsilon_i$

Selecting parameters

- The task is to select values for β_0 and β_1 which optimise our guess
- Our ε term captures how wrong we are i.e., the *residual*
- We want to minimise how wrong we are
- We can capture a metric of how wrong we are by summing the square of our residuals i.e., $\text{SSR} = \sum \varepsilon_i^2$
- Now we select values which will minimise this
- We could solve the problem numerically
- We can also use calculus

Equations to solve

$$\frac{\partial}{\partial \hat{\beta}_0} \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2 = -2 \sum (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)$$

$$\frac{\partial}{\partial \hat{\beta}_1} \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2 = -2 \sum x_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)$$

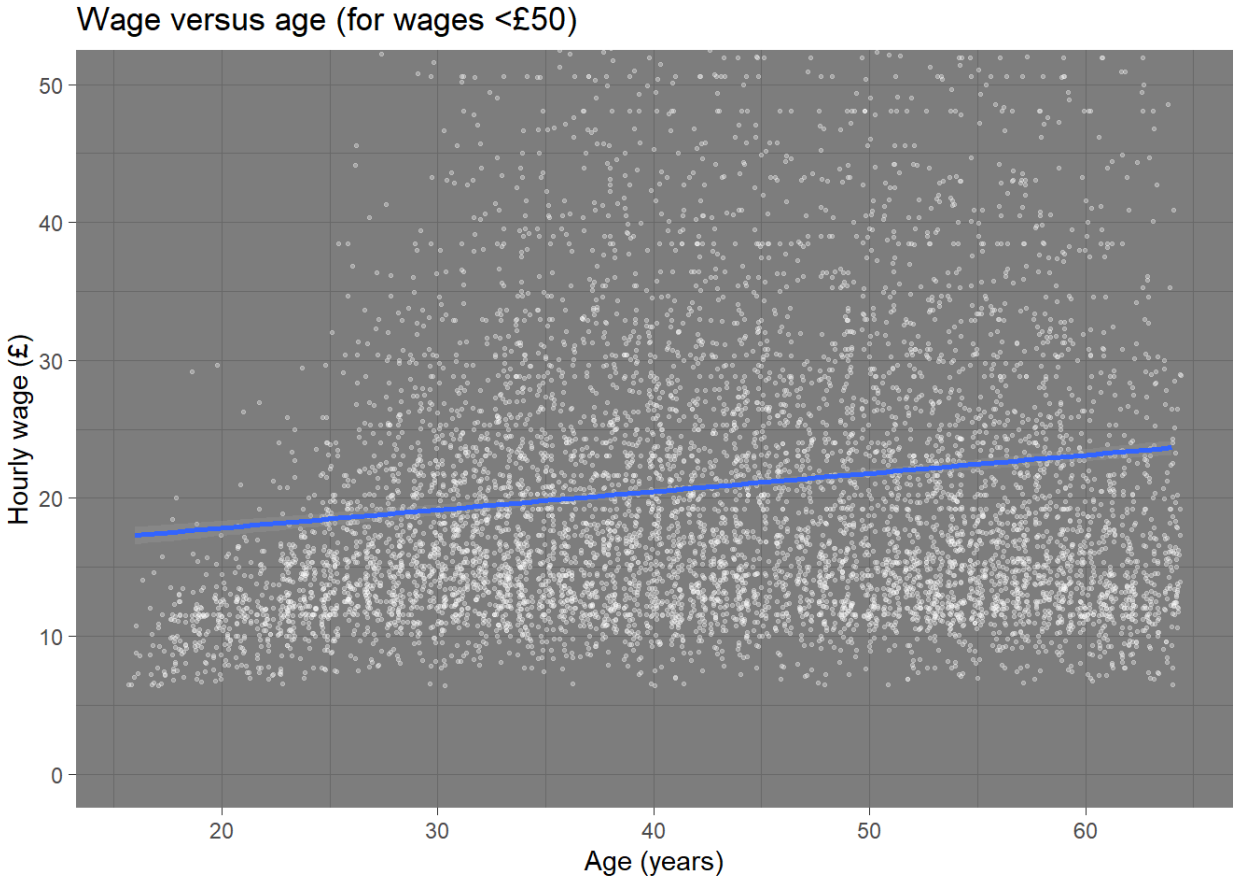
The solution

$$\hat{\beta}_1 = \frac{\frac{1}{n}\sum_{i=1}^{n} x_i y_i - \bar{x}\bar{y}}{\frac{1}{n}\sum_{i=1}^{n} x_i^2 - \bar{x}^2}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

The linear model

- We have already seen the solution for these parameters
- β_0 is the intercept from the linear model
- β_1 is the coefficient attached to age
- We can visualise what this looks like



The linear model

```
lm(HOURPAY ~ AGE, data=lfs) %>% summary()
```

```
##
## Call:
## lm(formula = HOURPAY ~ AGE, data = lfs)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -17.155  -7.615  -3.334   4.221  76.885
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## Coefficients:
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## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 11.43 on 7345 degrees of freedom
## Multiple R-squared:  0.01892,    Adjusted R-squared:  0.01879
## F-statistic: 141.6 on 1 and 7345 DF,  p-value: < 2.2e-16
```

Interpretation

- The intercept tells us the wage of a person who is zero years old
- This isn't a helpful interpretation
- In the linear model, we don't generally bother interpreting the intercept
- The coefficient attached to age tells us the association between age and wage
- Being a year older is associated with wages which are £0.13 higher
- The effect is statistically significant
- Variation in age accounts for 1.9% of variation in wages

Correlation and causation

Correlation is not causation

Cum hoc ergo propter hoc

- This is a well known logical fallacy
- All we show with these methods is an association
- We must be careful when writing up this kind of analysis
- Do not talk about proving things
- Do not talk about things causing things
- Talk about things being associated with other things

Summary

- We have seen some more of the magic of the linear model
- It can measure associations using categorical, ordinal and scale/interval independent variables
- It can test for statistical significance
- There are yet more wonders to behold